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A PROPOSAL REPORT
ON
AI-Based Job Recommendation Chatbot System

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List Of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
BCA	Bachelor in Computer Application
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DB	Database
HTML	Hypertext Markup Language
IT	Information Technology
JS	JavaScript
PHP	Hypertext Preprocessor
ML	Machine Learning
MySQL	My Structured Query Language
UI	User Interface
UX	User Experience

CHAPTER 1 INTRODUCTION

1.1 Introduction

In the modern digital era, job searching has become a complex and time-consuming process for students and job seekers. Many individuals struggle to find job opportunities that match their skills and qualifications. Traditional job portals provide numerous listings but lack personalized recommendations and intelligent guidance.

Another major challenge is that most job seekers, especially fresh graduates, are not fully aware of the skills required in the job market. They often apply for jobs without understanding whether they are a good fit or not. This leads to frustration, rejection, and confusion about career direction. As a result, there is a growing need for a system that not only recommends jobs but also guides users in improving their skills.

With the rapid growth of technology, Artificial Intelligence (AI) and Machine Learning (ML) are being used to solve real-world problems in smarter ways. These technologies can analyze user data, understand patterns, and provide personalized suggestions. In the context of job searching, AI can help match user skills with job requirements more accurately than traditional methods.

This project proposes an AI-Based Job Recommendation Chatbot System that interacts with users in a simple and user-friendly way. Instead of manually searching through job listings, users can simply chat with the system, enter their skills, or upload their resume. The chatbot then processes this information and provides relevant job recommendations instantly.

The system uses Machine Learning techniques such as TF-IDF and Cosine Similarity to analyze and compare user profiles with job descriptions. These techniques help in converting text data into numerical form and measuring how closely a user's skills match with job requirements. Additionally, the system identifies skill gaps and suggests areas where the user needs improvement, making it not just a recommendation system but also a guidance tool.

Overall, this project aims to simplify the job searching process and make it more efficient, personalized, and interactive. It is especially useful for students and fresh graduates who are just starting their careers. By combining chatbot technology with AI/ML techniques, the system provides a smart solution that bridges the gap between job seekers and job opportunities.

1.2 Problem Statement

In today's competitive job market, students and job seekers face significant difficulties in finding job opportunities that match their skills and qualifications. Most traditional job portals provide a large number of job listings, but they lack personalized recommendations, forcing users to manually search and filter jobs. This process is time-consuming, inefficient, and often frustrating. Additionally, many users, especially fresh graduates, are not fully aware of the skills required for specific job roles, which leads to confusion, repeated rejection, and lack of clear career direction.

Furthermore, existing systems do not provide proper guidance on skill improvement or identify the gap between user skills and job requirements. They also lack an interactive and user-friendly interface, making the job search experience complex and less engaging. Therefore, there is a need for an intelligent system that can automatically analyze user skills, recommend suitable jobs, and guide users in improving their qualifications. The proposed AI-Based Job Recommendation Chatbot System aims to solve these issues by using machine learning techniques to provide personalized recommendations through a simple chatbot interface.

1.3 Objectives

The main objectives of this project are:

- To develop an intelligent chatbot-based system that provides personalized job recommendations based on user skills.
- To analyze and match user skills with job requirements using machine learning techniques, proper using LLM.
- To identify skill gaps and provide provides guidance to users on the skills needed to improve for better job opportunities and career growth.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background study

In recent years, the rapid growth of digital technology has significantly transformed the way people search for jobs. Online job portals and recruitment platforms have made it easier for employers to post vacancies and for job seekers to explore opportunities. However, these platforms often present a large amount of information without proper personalization, making it difficult for users to find jobs that truly match their skills and interests.

Recommendation systems have emerged as an important solution to this problem. A recommendation system is a type of intelligent system that suggests relevant items to users based on their preferences, behavior, or profile data. In the context of job searching, recommendation systems analyze user skills and match them with job requirements to provide suitable job suggestions. This helps reduce the effort required by users to manually search through numerous job listings.

Machine Learning (ML) plays a key role in enhancing the performance of recommendation systems. ML techniques can process large amounts of data, identify patterns, and make predictions. In this project, techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) are used to convert textual data like skills and job descriptions into numerical form, which can be processed by algorithms.

Another important concept used in this system is Cosine Similarity. It is a mathematical method used to measure the similarity between two vectors. In this case, it is used to determine how closely a user's skills match with the requirements of a job. A higher similarity score indicates a better match, allowing the system to rank and recommend jobs more accurately.

Chatbot technology is also an important part of modern systems. Chatbots provide a conversational interface that allows users to interact with the system in a simple and natural way. Instead of navigating complex websites, users can directly communicate with the chatbot by typing their skills or uploading their resume. This improves user experience and makes the system more accessible.

In addition, resume parsing has become a useful technique in job recommendation systems. It involves extracting useful information such as skills, education, and experience from a resume document. This helps automate the process of skill identification and improves the accuracy of recommendations.

Overall, the combination of recommendation systems, machine learning techniques, and chatbot interfaces provides a powerful solution for improving the job search process. These technologies work together to deliver personalized, efficient, and user-friendly job recommendations, making them highly relevant for modern job seekers.

2.2 Literature review

The integration of artificial intelligence and machine learning in job recommendation systems has gained significant attention in recent years. With the increasing volume of job postings and user data, intelligent systems are required to provide personalized and efficient recommendations. According to various studies, machine learning techniques such as content-based filtering, TF-IDF, and classification models are widely used to match user skills with job descriptions based on textual data [1]. These techniques enable systems to analyze large datasets and provide relevant job suggestions by identifying patterns and similarities.

Several researchers have emphasized the importance of text processing and similarity measurement in improving recommendation accuracy. Techniques such as TF-IDF are commonly used to convert textual information into numerical vectors, allowing systems to process and compare user profiles effectively. Cosine similarity, in particular, is widely adopted as a reliable method for measuring the similarity between user skills and job requirements. Studies show that this approach improves the precision of job recommendations by ranking results based on similarity scores [2].

Natural Language Processing (NLP) also plays a vital role in job recommendation systems, especially in resume parsing and skill extraction. By applying NLP techniques such as tokenization, stop-word removal, and stemming, systems can extract meaningful information from resumes and job descriptions. Research indicates that integrating NLP with machine learning models enhances the performance of job matching systems by improving data preprocessing and feature extraction [3].

In recent years, advanced approaches such as semantic analysis and word embeddings have been introduced to further improve recommendation quality. Models like Word2Vec and BERT capture contextual relationships between words, allowing systems to understand the meaning of skills and job requirements more effectively. Although these models provide higher accuracy, studies suggest that they require more computational resources compared to traditional approaches like TF-IDF [4].

Furthermore, chatbot-based interfaces have been increasingly incorporated into recommendation systems to improve user interaction. Chatbots provide a conversational platform where users can input their skills and receive instant job recommendations. Research shows that chatbot integration enhances user experience by simplifying the interaction process and making systems more accessible, especially for non-technical users [5].

Despite these advancements, many existing job recommendation systems still lack a balance between accuracy, simplicity, and cost-effectiveness. Advanced models often require complex infrastructure, while simpler systems may lack intelligence and personalization. Therefore, there is a need for a practical and efficient solution that combines machine learning techniques with user-friendly interfaces.

This project focuses on developing an AI-based job recommendation chatbot system that integrates TF-IDF, cosine similarity, and NLP techniques to provide personalized job suggestions. By combining intelligent algorithms with a simple chatbot interface, the system aims to deliver an effective and accessible solution for job seekers, particularly students and fresh graduates.

CHAPTER 3: METHODOLOGY

3.1 Requirement identification

The AI-Based Job Recommendation Chatbot System requires a platform that allows users to interact through a chatbot interface to input their skills or upload resumes. The system should be able to process textual data, extract relevant skills, and match them with job requirements stored in a database. It must include a machine learning component to calculate similarity between user profiles and job descriptions using techniques such as TF-IDF and cosine similarity. Additionally, the system should support features like job filtering based on location and salary, skill gap identification, and an admin panel for managing job data. The system should be user-friendly, efficient, and capable of providing accurate and timely job recommendations.

3.1.1 Requirement analysis

Functional Requirements

a) User Interaction (Chatbot)

- Users should be able to interact with the chatbot through a simple interface.
- Users should be able to enter their skills manually.
- The system should respond with job recommendations based on user input.
- Users should be able to ask multiple queries continuously.

b) Resume Management

- Users should be able to upload their resume in PDF format.
- The system should extract text from the uploaded resume.
- The system should identify and extract relevant skills from the resume.
- Extracted skills should be used for job recommendation.

c) Job Recommendation System (AI/ML)

- The system should match user skills with job requirements.
- The system should use TF-IDF to convert text into numerical vectors.
- The system should use cosine similarity to calculate similarity scores.
- The system should rank jobs based on similarity scores.
- The system should display top matching jobs to users.

d) Skill Gap Analysis

- The system should compare user skills with job requirements.
- The system should identify missing skills.
- The system should suggest skills required for better job opportunities.

e) Job Filtering System

- Users should be able to filter jobs based on location.
- Users should be able to filter jobs based on salary.
- The system should display filtered job results accordingly.

f) Job Management (Admin)

- The admin should be able to log in securely.
- The admin should be able to add new job listings.
- The admin should be able to update existing job details.
- The admin should be able to delete job listings.
- The system should display jobs based on admin input.

g) Data Storage

- The system should store user data, job data, and match scores in the database.
- The system should retrieve job data for recommendation.

Non-Functional Requirements

a) Usability

- The system should have a simple and user-friendly chatbot interface.
- It should be easy to use for students and fresh graduates.
- The interface should be responsive on different devices.

b) Performance

- The system should provide fast responses to user queries.
- The recommendation process should be efficient.
- The system should handle multiple users simultaneously.

c) Security

- The system should provide secure admin authentication.
- User data and resume data should be handled securely.
- Sensitive data should be protected from unauthorized access.

d) Scalability

- The system should support future enhancements like advanced AI models.
- It should allow integration with external job portals.
- The system should be designed in a modular way.

e) Reliability

- The system should provide accurate job recommendations.
- It should function consistently without errors.
- The system should be available for continuous use.

f) Data Integrity

- The system should ensure correct storage of job and user data.
- Data should remain consistent during updates and processing.
- The system should avoid duplication or loss of data.

3.2 Feasibility Study

The feasibility study is conducted to evaluate the practicality and viability of the proposed AI-Based Job Recommendation Chatbot System. It helps determine whether the system can be successfully developed and implemented within available resources and constraints.

3.2.1 Technical Feasibility

The proposed system is technically feasible as it uses widely available and well-supported technologies such as PHP, MySQL, HTML, CSS, JavaScript, and Python for machine learning. The AI/ML component uses simple and efficient techniques like TF-IDF and cosine similarity, which do not require high computational power. Development tools such as XAMPP and libraries like scikit-learn and PDF parsers are easily accessible and suitable for implementation. Therefore, the system can be developed using existing technical knowledge and resources.

3.2.2 Economic Feasibility

The system is economically feasible as it requires minimal cost for development and deployment. All the tools and technologies used in the project are open-source and freely available. The development can be carried out on a standard personal computer without the need for expensive hardware or software. Since there are no significant operational or maintenance costs, the project is cost-effective and suitable for academic purposes.

3.2.3 Operational Feasibility

The system is operationally feasible as it is designed to be simple and user-friendly. Users can interact with the chatbot easily without requiring advanced technical knowledge. The admin panel allows easy management of job data. The system provides quick and relevant job recommendations, improving user experience and efficiency in job searching. Hence, it can be effectively used by students and job seekers.

3.3 High Level Design of System

The high-level design of the system provides an overview of how different components interact to achieve the desired functionality. The proposed system follows a modular and layered architecture that integrates frontend, backend, database, and machine learning components to deliver an intelligent job recommendation chatbot system efficiently. The system is designed to process user input, analyze skills, and generate personalized job recommendations through an interactive chatbot interface.

3.3.1 Methodology and Working Mechanism of the Proposed System

The proposed system follows an Agile methodology, where development is carried out in small and iterative phases. Instead of developing the entire system at once, individual modules such as user interaction (chatbot), resume processing, job recommendation engine,

and admin management are implemented step by step. After each phase, testing and improvements are performed, allowing flexibility and continuous refinement. This approach helps in identifying errors early, incorporating feedback, and ensuring smooth integration of machine learning components with the web application. The overall working mechanism of the system is explained below:

User Interaction (Chatbot Interface):

Users interact with the system through a chatbot interface by entering their skills or queries. The chatbot provides a simple and user-friendly way to communicate with the system.

User Input / Resume Upload:

Users can either manually input their skills or upload their resume in PDF format. The system accepts and processes both types of input for analysis.

Resume Parsing and Skill Extraction:

If a resume is uploaded, the system extracts textual information using parsing techniques. Relevant skills, education, and keywords are identified using Natural Language Processing (NLP) methods.

Data Processing and Feature Extraction:

The extracted skills and job descriptions are preprocessed using techniques such as tokenization, stop-word removal, and text normalization. TF-IDF is applied to convert textual data into numerical vectors.

Job Matching Process:

The processed user data is compared with job descriptions stored in the database. Cosine similarity is used to calculate similarity scores between user skills and job requirements.

Job Recommendation Generation:

Based on similarity scores, the system ranks available jobs and selects the most relevant ones. The top matching jobs are recommended to the user through the chatbot interface.

Skill Gap Analysis:

The system identifies missing skills by comparing user profiles with job requirements. It suggests skills that users need to improve for better job opportunities.

Result Display and Interaction:

Recommended jobs and skill suggestions are displayed to the user via the chatbot. Users can continue interacting with the system by asking further queries.

Admin Management and Data Handling:

The admin manages job listings by adding, updating, or deleting job data. The system stores user data, job data, and recommendation results in the database for efficient retrieval and processing.

3.3.2 Flowchart

A flowchart is a graphical representation that illustrates the step-by-step workflow of a system using standard symbols and arrows to show the flow of processes. In the AI-Based Job Recommendation Chatbot System, the flowchart represents the complete working mechanism starting from user interaction with the chatbot, where users input their skills or upload a resume, followed by data processing, skill extraction, and application of machine learning techniques such as TF-IDF and cosine similarity for job matching. It further shows how the system generates and displays relevant job recommendations along with skill gap suggestions. This visual representation helps in understanding the overall system logic, data flow, and interaction between different components in a clear and simplified manner.

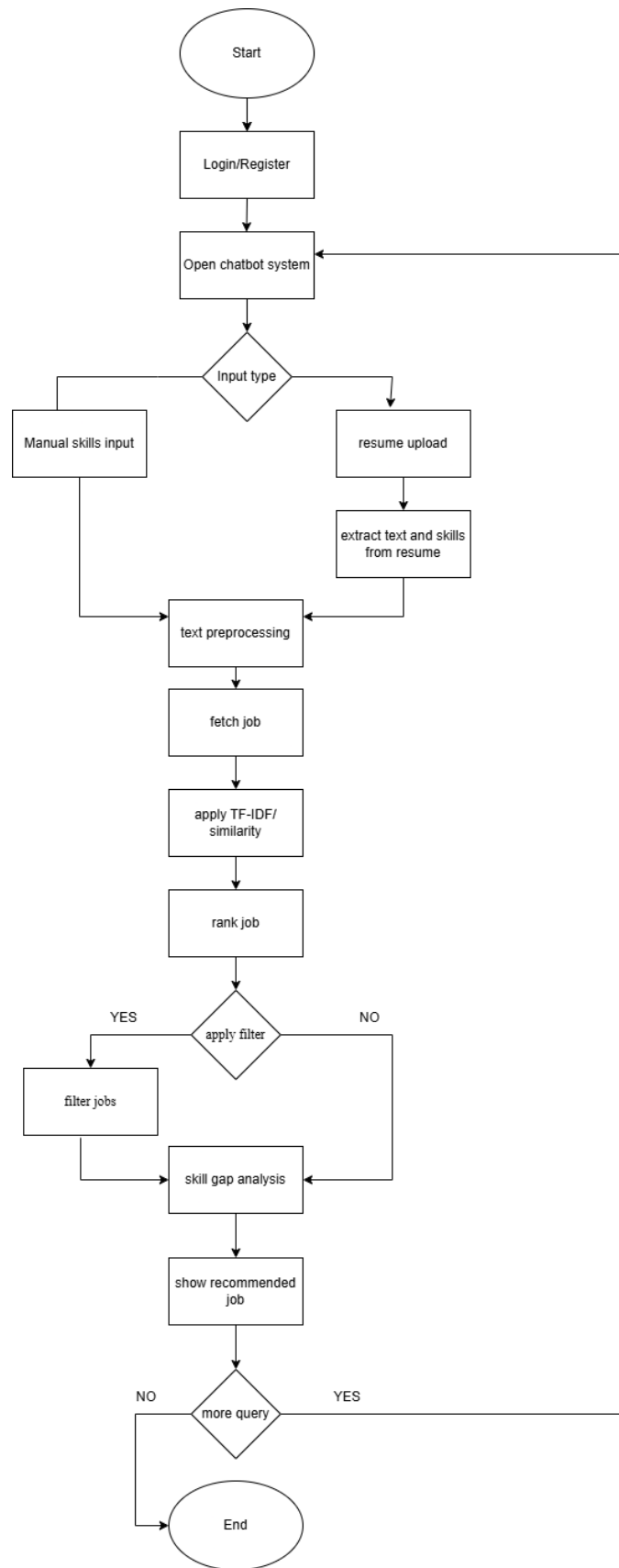


Figure 1: Flowchart

3.3.3 Data modeling ER

An Entity Relationship (ER) diagram is a visual representation of the database structure of a system that shows the entities, their attributes, and the relationships among them. In the AI-Based Job Recommendation Chatbot System, the ER diagram helps illustrate how different entities such as user, resume, skills, jobs, recommendations, and admin are connected within the database. It provides a clear view of how user information is stored, how job data is managed, and how recommendations are generated based on the relationship between user skills and job requirements. This diagram is useful for understanding the logical design of the database and ensures proper organization, consistency, and efficient management of system data.

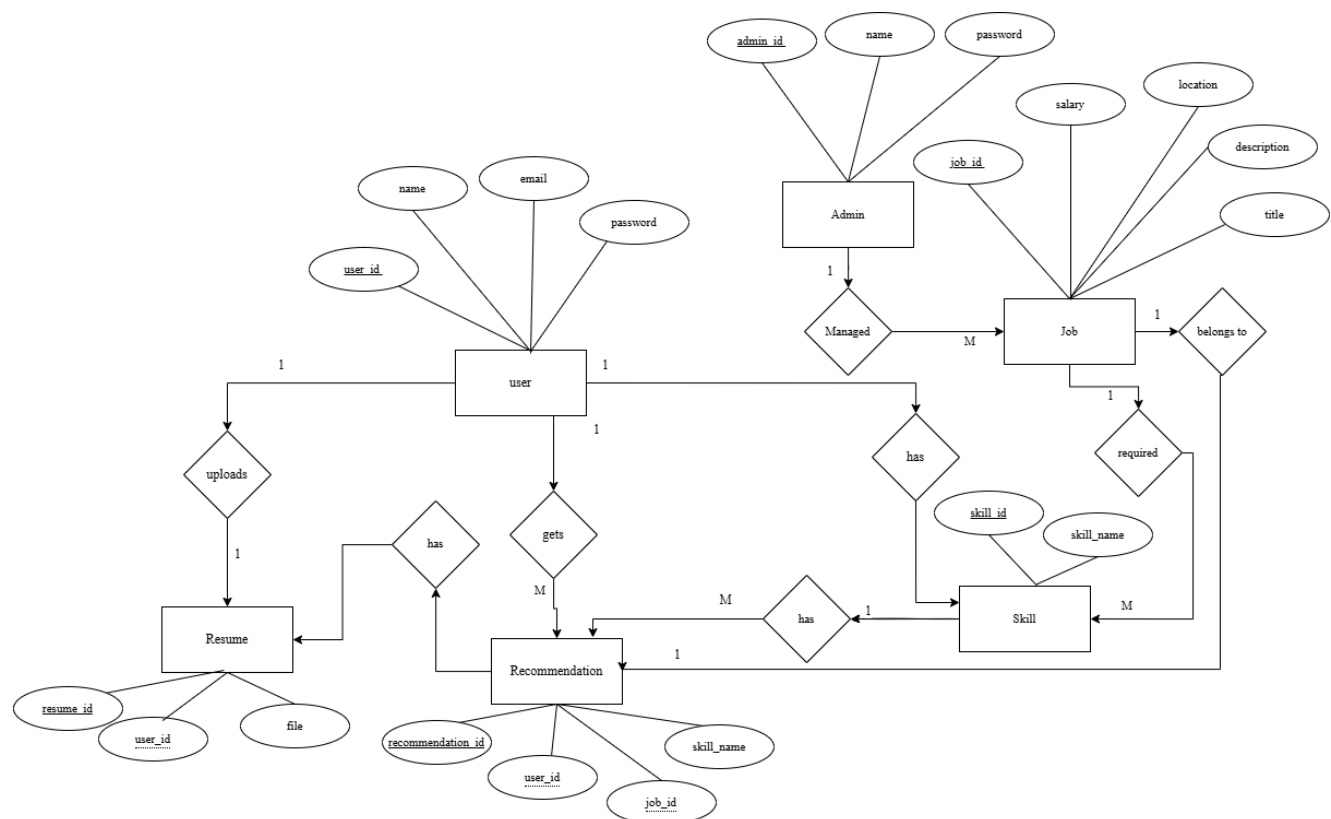


Figure 2:Er diagram

3.4 Gantt Chart

The Gantt chart represents the timeline and phases of the project development. It provides a structured schedule to ensure that each stage of the project is completed within the allocated time. The project is planned to be completed over a period of 12 weeks, covering all major phases from requirement analysis to final testing and documentation.

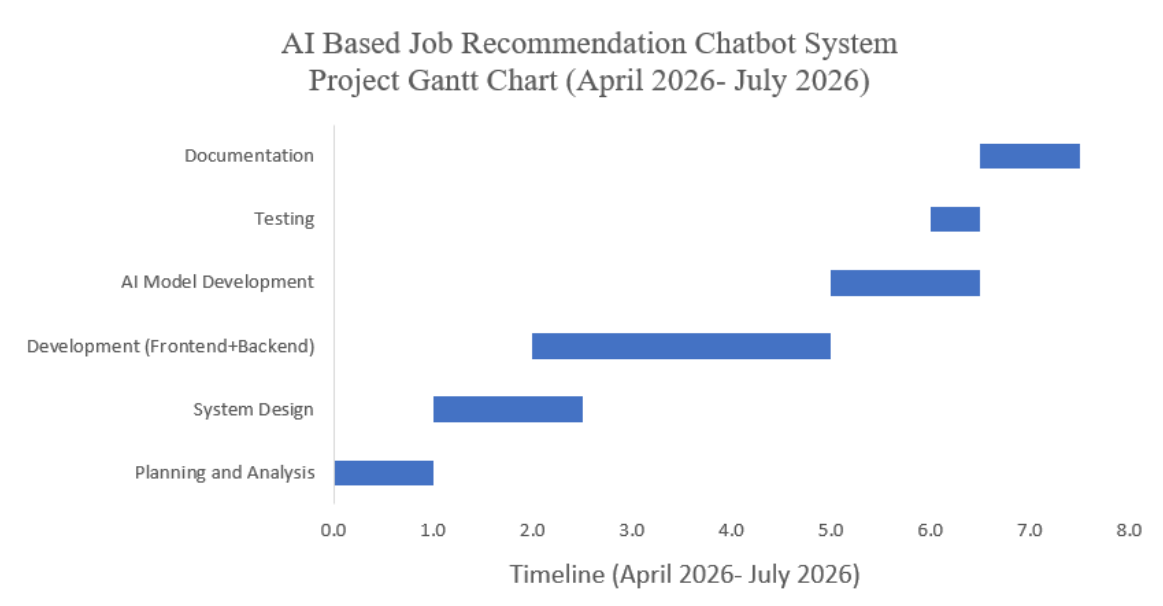


Figure 3: Gantt chart

CHAPTER 4: EXPECTED OUTCOMES

The expected outcome of this project is the development of a fully functional, web-based AI-Based Job Recommendation Chatbot System that provides personalized job suggestions using machine learning techniques. The system is designed to simplify the job searching process for users while helping them identify suitable career opportunities based on their skills and qualifications. By analyzing user input and job descriptions, the system will generate accurate job recommendations and provide guidance for skill improvement, making it more effective than traditional job portals.

a) Intelligent Job Recommendation System

The system analyzes user skills and compares them with job requirements to recommend the most relevant job opportunities. For example, if a user has skills in programming or web development, the system will suggest jobs related to those fields. This makes the system smarter and more personalized than traditional job search platforms.

b) Skill Gap Analysis

The system identifies missing skills by comparing user profiles with job requirements. It suggests important skills that users need to improve in order to qualify for better job opportunities. This helps users understand their weaknesses and plan their career growth effectively.

c) Data-Driven Decision Support

The system provides useful insights such as matching scores, required skills, and job trends. This helps users make informed decisions about which jobs to apply for and which skills to develop further.

d) Overall Benefits

This system makes the job searching process more efficient, interactive, and intelligent. It improves user experience through a chatbot interface and reduces the time and effort required to find suitable jobs, while helping users enhance their skills and career opportunities.

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